

Appendix E

Non-equilibrium work and detailed balance

The relation between free-energy differences and (non-equilibrium) work described in section 8.7 holds for all protocols or equations of motion that are markovian and satisfy detailed balance for every step.

To see what that means, consider a process where work is performed by changing a control parameter Λ . For instance, Λ might be the system's volume, or it might be a parameter in the Hamiltonian of the system. We denote the original/final value of Λ as Λ_0/Λ_K (we use the index K to keep the subsequent notation consistent).

The protocol to change Λ from Λ_0 to Λ_K can be decomposed in two types of elementary steps: during the first, all phase-space coordinates of the system (Γ) are kept fixed, and Λ is changed by an amount $\Delta\Lambda_i$, where i labels the step; if there are K such steps, then $i = \{1, 2, \dots, K\}$. Note that steps that only change Λ are deterministic. However, there is also a second type of step in the protocol, namely one where Λ is kept fixed, but the system is allowed to evolve by its natural dynamics, exchanging energy with a thermostat. Examples of the evolution at constant Λ are sequences of one or more Monte Carlo moves, or of one or more time steps in a constant-temperature MD simulation.¹ In the language of Crooks [389], we denote the energy exchanged with the “reservoir” (i.e., the thermostat) by Q : it can be interpreted as the heat delivered to the system by the reservoir. Because of detailed balance, the ratio of the probability of the system to evolve at constant Λ (say Λ_i) from phase-space coordinate Γ to Γ' to the probability of the reverse move, is given by

$$\frac{P(\Gamma \rightarrow \Gamma'; \Lambda_i)}{P(\Gamma' \rightarrow \Gamma; \Lambda_i)} = e^{-\beta \Delta E(\Lambda_i)}, \quad (\text{E.1.1})$$

where $\Delta E(\Lambda_i) \equiv E(\Gamma'; \Lambda_i) - E(\Gamma; \Lambda_i)$.

When we change Λ at constant Γ , we perform work on the system. We denote the work associated with a change Λ from Λ_{i-1} to Λ_i at fixed Γ_i by w_i . The total work W done on the system as Λ is increased from Λ_0 to Λ_K , is then equal to $W = \sum_{i=1}^K w_i$.

¹ The argument can be generalized to the case where we also exchange volume or particles with a reservoir, but here we consider the simplest case.

If the time evolution of the system is Markovian, we can write the probability to evolve from Γ_0 at Λ_0 to Γ_K at Λ_K as

$$\prod_{i=0}^{K-1} [P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i) \times 1] .$$

We included a factor one in every step to indicate that changing Λ at constant Γ is deterministic. In what follows, we leave out this trivial factor. We can also write down the probability to evolve from Γ_K at Λ_K to Γ_0 at Λ_0 along the same path, as

$$\prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i) .$$

Because of Eq. (E.1.1), we have

$$\prod_{i=0}^{K-1} P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i) = \prod_{i=K-1}^0 \left[P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i) e^{-\beta \Delta E(\Lambda_i)} \right]. \quad (\text{E.1.2})$$

Note that $\prod_{i=K-1}^0 e^{-\beta \Delta E(\Lambda_i)}$ is equal to $e^{-\beta Q(\{\Gamma\})}$, where Q is the total energy transferred from the reservoir for a sequence of states $\{\Gamma\} \equiv \Gamma_0, \Gamma_1, \dots, \Gamma_K$. Note that Q is *not* equal to $E(\Gamma_K; \Lambda_K) - E(\Gamma_0; \Lambda_0)$ because the energy of the system is also changed by performing work:

$$E(\Gamma_K; \Lambda_K) - E(\Gamma_0; \Lambda_0) = W(\{\Gamma\}) + Q(\{\Gamma\}) , \quad (\text{E.1.3})$$

which can be viewed as a microscopic version of the First Law of Thermodynamics. Note, in particular, that W and Q are path-dependent, but their sum is not. If we sample initial conditions from a Boltzmann distribution, then we can express the average of $e^{-\beta W}$ as

$$\overline{e^{-\beta W}} = \sum_{\Gamma_0, \dots, \Gamma_K} P_B(\Gamma_0, \Lambda_0) \prod_{i=0}^{K-1} P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i) e^{-\beta W(\{\Gamma\})} , \quad (\text{E.1.4})$$

where $P_B(\Gamma_0, \Lambda_0) = \exp(-\beta[E(\Gamma_0; \Lambda_0) - F(\Lambda_0)])$, and the Helmholtz free energy F is, as usual, given by

$$\beta F(\Lambda) = -\ln \sum_{\Gamma} e^{-\beta[E(\Gamma; \Lambda)]} . \quad (\text{E.1.5})$$

Using Eq. (E.1.2), we can write

$$\frac{\prod_{i=0}^{K-1} P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i)}{\prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i)} = e^{-\beta Q(\{\Gamma\})} \quad (\text{E.1.6})$$

or, using the definition of the Boltzmann weights:

$$\frac{P_B(\Gamma_0, \Lambda_0) \prod_{i=0}^{K-1} P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i)}{P_B(\Gamma_K, \Lambda_K) \prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i)} = e^{-\beta Q(\{\Gamma\})} e^{+\beta[\Delta E - \Delta F]}, \quad (\text{E.1.7})$$

where $\Delta E \equiv E(\Gamma_K; \Lambda_K) - E(\Gamma_0; \Lambda_0)$, and $\Delta F \equiv F(\Lambda_K) - F(\Lambda_0)$. We can then write:

$$\begin{aligned} P_B(\Gamma_0, \Lambda_0) & \prod_{i=0}^{K-1} P(\Gamma_i \rightarrow \Gamma_{i+1}; \Lambda_i) e^{-\beta W(\{\Gamma\})} \\ &= P_B(\Gamma_K, \Lambda_K) \prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i) e^{-\beta[Q(\{\Gamma\}) + W(\{\Gamma\}) - \Delta E]} e^{-\beta \Delta F} \\ &= P_B(\Gamma_K, \Lambda_K) \prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i) e^{-\beta \Delta F}, \end{aligned} \quad (\text{E.1.8})$$

where we have used Eq. (E.1.3). It then follows that Eq. (E.1.4) can be written as

$$\overline{e^{-\beta W}} = e^{-\beta \Delta F} \sum_{\Gamma_0, \dots, \Gamma_K} \prod_{i=K-1}^0 P(\Gamma_{i+1} \rightarrow \Gamma_i; \Lambda_i) = e^{-\beta \Delta F}, \quad (\text{E.1.9})$$

where the last equality follows from the fact that all transition probabilities are normalized.